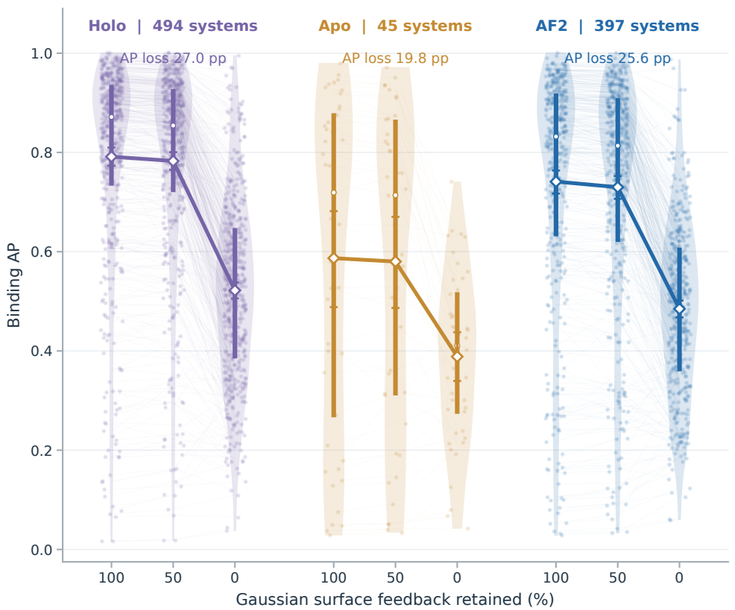
Gaussian surface support for accurate binding-site prediction

A frozen model, controlled pathway interventions, and sequence-audited conformational pairs

COMPUTATIONAL DEPENDENCE / DOSE RESPONSE / SITE PRIORITIZATION

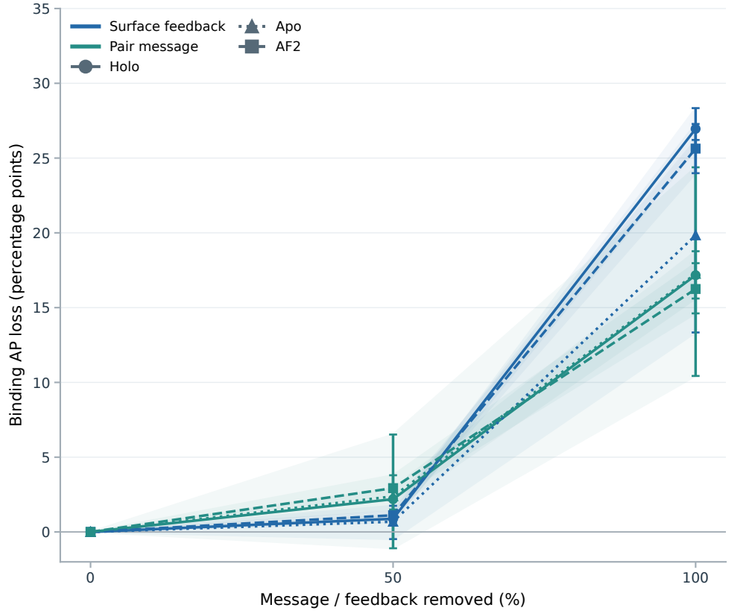
a The surface pathway supports prediction

Paired systems; distributions + IQR; diamonds: mean and 95% CI



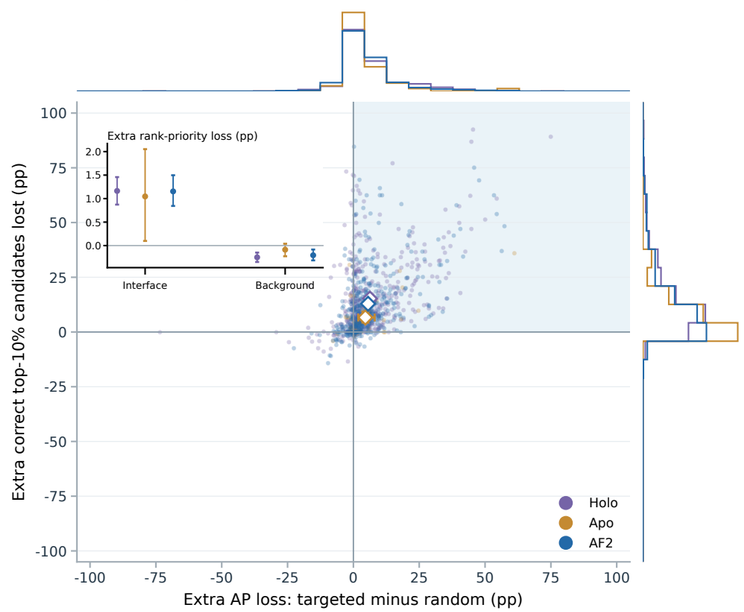
b Dependence has a graded response

Two intervention sites; three observed doses; shared system bootstrap



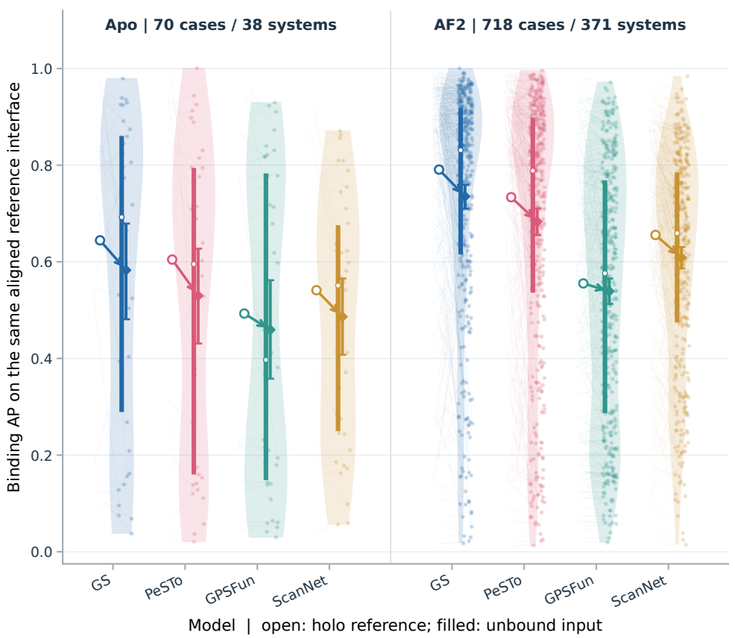
c Local pair removal disrupts priority

High-score regions vs same-count random surface points; three seeds



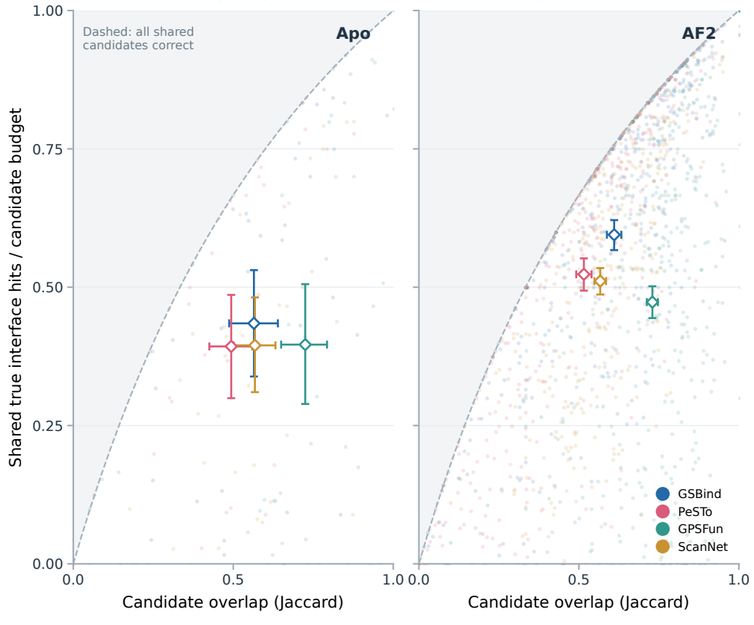
d Accurate interfaces survive input changes

One line per system; common positions and fixed holo labels



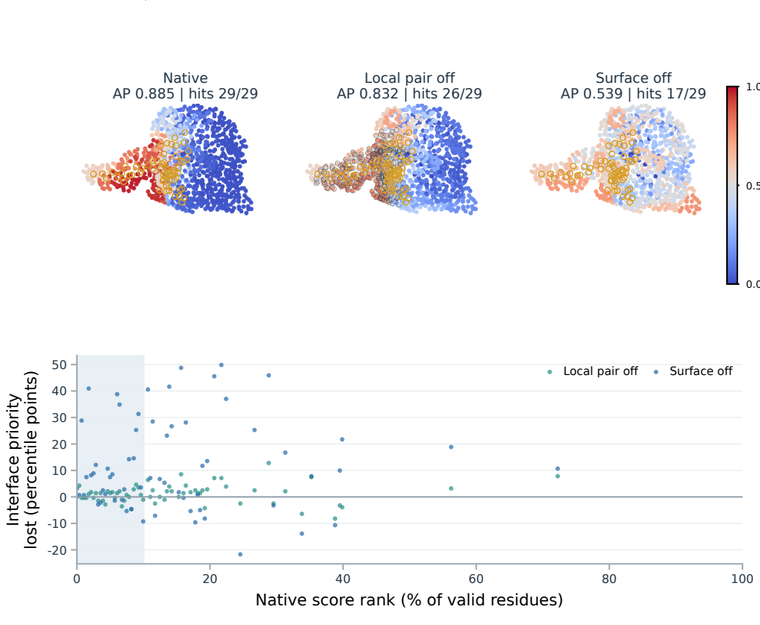
e Preserve correct candidates, not just overlap

Fixed top-10% budgets; system clouds + paired mean uncertainty



f Gaussian support localizes the interface

4ml9 R | gold: true interface; surface color: nearest-residue score



Interventions: 988 holo / 84 apo / 790 AF2 cases. Transfer: 70 apo / 718 AF2 uniquely aligned cases. Means are system-equal; 95% CIs use 10,000 system resamples.